Direct FRAMES–Quiroga Comparison Setup

Implementation and experiment-readiness report · 4 October 2026

> **Outcome:** the paired comparison is implemented, committed locally on Elite, and validated. The exact n=2 launcher-equivalent smoke completed for both methods. No Ruche job or production sweep has been submitted.

Item	Status
Local source revision	`3bf36c4f47ac2c637df9fe3e1d9db0d8173484e2`
Focused comparison tests	8 passed, 0 failed
Quiroga/FRAMES regression tests	96 passed, 0 failed
Full repository suite	525 passed, 55 skipped, 0 failed
Elite n=2 paired smoke	Both methods complete; matched 256 nominal row accesses each; artifacts verified
GitHub / origin	Not updated: push requires explicit authorization for code egress
Ruche	Script and resource request staged locally; no checkout created and no job submitted

What is now a direct comparison

Every primary run invokes printed adaFGD and stochastic FRAMES together on the same Quiroga-style global-probe / generalized 2d-outcome-POVM dataset. They receive the same active rows, observations, truth factor, rank, and serialized initial factor.

The portable penalty parameter is the per-row coefficient α. For m active rows:

$$L_\alpha(U) = (1 / 2m)\, \|A_S(UU^H) - y_S\|^2 + \alpha\, \|\mathrm{Tr}_{out}(UU^H) - I\|^2$$

printed adaFGD: $\lambda = m\alpha$ FRAMES: $\beta = 1 / (2\alpha)$

This makes the printed summed objective exactly m times the FRAMES mean-loss objective. The primary grid is

shared across methods:

α	Equivalent fixed FRAMES β
0.5	1
0.05	10
0.005	100
0.0005	1,000
0.00005	10,000

Earlier n=4 held-out artifacts used different penalties for the two methods (adaFGD β=100, FRAMES β=1,000). They remain preserved as exploratory evidence but are not counted as the new primary comparison.

Fairness and statistical unit

- **Budget:** equal logical active-row epochs and aligned checkpoints. These are not compute-equivalent: adaFGD also pays for a spectral-norm numerator, while FRAMES uses stochastic batches.
- **Primary scientific unit:** an independent channel/data instance—not a stochastic sampling seed.
- **Aggregation:** sampling replicates are averaged within one initialization; initializations are averaged within one channel; channels are then weighted equally.
- **Gate estimand:** the primary success rate is the nested mean of Boolean gate outcomes. “All replicates pass” is retained only as a robustness diagnostic, because it changes with the number of stochastic replicates.
- **Domain disclosure:** FRAMES uses an operator-norm ball; printed adaFGD is unconstrained. The runner verifies that the truth and initial factor lie inside the chosen FRAMES ball.

> The baseline is the literal adaptive update printed by Quiroga and Kyrillidis, with project-calibrated constants where the paper is underspecified. At n=4 this is a Quiroga-style higher-qubit extension, not a claim of exact reproduction of an unpublished n=4 setup.

Recovery gate and staged ladder

A noiseless endpoint passes only when all three conditions hold:

Metric	Threshold
Process fidelity	≥ 0.99
Full-design measurement RMSE	$\leq 10^{-6}$

$\|\mathrm{Tr}_{out}(UU^H) - I\|_F / \sqrt{d}$ $\leq 10^{-3}$

1. **n=2 full design:** all 128 rows.
2. **n=2 published-scale subset:** 96 shared training rows, with the gate evaluated on all 128 rows at the endpoint.
3. **n=4 full design:** all 8,192 rows, only after n=2 accounting and archive checks pass.

Calibration uses fresh, role-separated seeds and successive budgets of 400, 2,000, then at most 10,000 row epochs for n=2 and 20,000 for n=4. The printed-adaFGD η grid is 10^{-4}, 3×10^{-4}, 10^{-3}, 3×10^{-3}, 10^{-2}, and 3×10^{-2}. Hyperparameters are frozen before fresh validation and final held-out channels.

Elite smoke evidence

The exact n=2 full-design smoke configuration intended for Ruche was run locally for two logical epochs. Both optimizers completed and were charged 256 active-row accesses. It is a plumbing check, not a recovery result; neither method is expected to pass after two epochs.

Artifact	SHA-256
pair.npz	`423e4e4723c28d5773362864de14471e7e23236ff3478667b77e3d68d283095d`
pair.json	`25accd1752caf2b47848947c81be04a3c122db2952da4f5dae8fd7ed447cca4b`

Ruche-ready smoke

Setting	Value
Partition / time	`cpu_short` / 00:20:00
Resources	1 node, 1 task, 2 CPUs, 1 GiB RAM
Environment	`anaconda3/2023.09-0/none-none`, Conda environment `qpt`
Immutable checkout target	`/gpfs/workdir/silvetian/frankwolfe-quiroga-3bf36c4`
Output target	`/gpfs/workdir/silvetian/qpt_quiroga_direct_3bf36c4_20261004_01/smoke`

The batch script pins the full commit, requires a clean worktree, keeps results outside the checkout, records software/Slurm provenance, refuses overwrite, and exits nonzero if either method fails.

> **Not yet synchronized:** the commit is one revision ahead of `origin/stochastic-frames`. The attempted push was blocked pending explicit authorization. Until that is approved, the Ruche checkout and scheduler smoke should not be created from an unshared revision.

Known limitations before paper production

- Current paired sidecars rerun deterministic adaFGD for each FRAMES sampling seed. Aggregation verifies and collapses identical scientific endpoints, but the production launcher should compute/reuse adaFGD once.
- The built-in aggregator validates JSON sidecars, not the associated NPZ arrays and checksums. A separate archive validator is required before paper use.
- If FRAMES raises mid-run, the current core retains the shared initial state, not the last finite FRAMES iterate.
- Fixed method order makes recorded optimizer times descriptive only. Controlled speed claims need fresh processes, balanced/randomized order, and hardware accounting.
- For the 96-row subset, full-design recovery is endpoint-only until every checkpoint factor is retained or reevaluated on all 128 rows.

Immediate next move

Authorize the push of commit `3bf36c4` to `origin/stochastic-frames`. Then create the commit-specific Ruche checkout, transfer this setup bundle, submit exactly one scheduler smoke, inspect its JSON/NPZ/logs and `seff`, and only then open the bounded n=2 calibration array. No n=4 production work should start before that n=2 pipeline passes.

Setup directory: `/home/tony/qpt_quiroga_direct_20261004_01`

Protocol: `/home/tony/frankwolfe/docs/qpt_quiroga_direct_comparison.md`

Primary paper reference reviewed: arXiv:2312.01311